CPRG 211 - Object-Oriented Programming 2

Description
This course builds on the concepts and languages you have learned previously. You will explore object-oriented programming concepts using an industry recognized language. Topics include object-oriented design concepts, polymorphism, inheritance, exception handling, serialization, and database connectivity.

3 Credits

Time Guidelines
The standard instructional time for this course is 75 hours.

Prerequisites

- CPRG 216

Course Assessment

Assignments	60%
Project(s)	40%
Total:	100%

Use of Turnitin

The instructor may submit student work in this course to Turnitin's text-matching software program, to help assess the academic integrity of student work in this course. Turnitin results may be considered as one piece of evidence in academic misconduct hearings. Turnitin is an American company that keeps no unencrypted student identity data in the United States unless the students themselves choose to share this information in their submission. Students should limit their sharing of personally identifiable information by not including their names and student ID numbers within the text body of submitted assignments. SAIT has carefully reviewed this company's data management procedures.

Other Course Information
Learner Engagement:

In order to be successful, the learner is expected to be engaged in learning activities for a total of 9 to 12 learning hours per course per week, which includes both in-class and out-of-class time.

Course Learning Outcomes
1. Apply object-oriented programming principles and class relationships in the design of projects.

Objectives:

1.1 Use programming language syntax in an integrated development environment (IDE).

1.2 Describe classes and objects.

1.3 Apply instance and static variables and methods.

1.4 Define abstraction and encapsulation.

1.5 Use a Unified Modelling Language (UML) class diagram.

1.6 Explain private, public, protected and default access modifiers.

1.7 Describe software development coding standards.

1.8 Explain packages and namespaces.

1.9 Use executable files.

1.10 Describe inheritance and polymorphism.

1.11 Describe the advantage of code reuse through inheritance and polymorphism.

1.12 Solve problems using inheritance and polymorphism.

1.13 Describe abstract classes and Interfaces and the difference between them.

2. Demonstrate exception handling techniques and serialization

Objectives:

2.1 Describe exceptions.

2.2 Compare errors and exception types.

2.3 Demonstrate how to throw exceptions.

2.4 Apply try blocks, catch blocks and throws.

2.5 Apply user-defined exceptions.

2.6 Demonstrate how to develop applications with exception handling.

2.7 Apply serialization and deserialization

2.8 Implement a serializable interface for enhance object portability and persistence.

3. Summarize the software testing methodology.

Objectives:

3.1 Describe testing in the software development cycle.

3.2 Define types of testing.

3.3 Explain the importance of testing in software development.

3.4 Review industry standard testing frameworks.

3.5 Apply a testing framework using a test-driven development methodology.

4. Apply singly linked lists (SLL) as a data structure mechanism to manage and manipulate data.

Objectives:

4.1 Define data structures.

4.2 Describe a singly linked list (SLL).

4.3 Compare array lists and linked lists.

4.4 Describe linked list types.

4.5 Define an interface for an SLL.

4.6 Demonstrate how to develop an SLL.

5. Connect a software program to a database.

Objectives:

5.1 Describe the process of connecting a program to a database.

5.2 Demonstrate how to manage data from a database using a program.

5.3 Apply auto commit, commit and rollback statements to a database.

SAIT Policies and Procedures:
Students should familiarize themselves with SAIT's Policies and Procedures, including but not limited to those in the Academic Student, Conduct, and Health, Safety, and Environment domains.

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